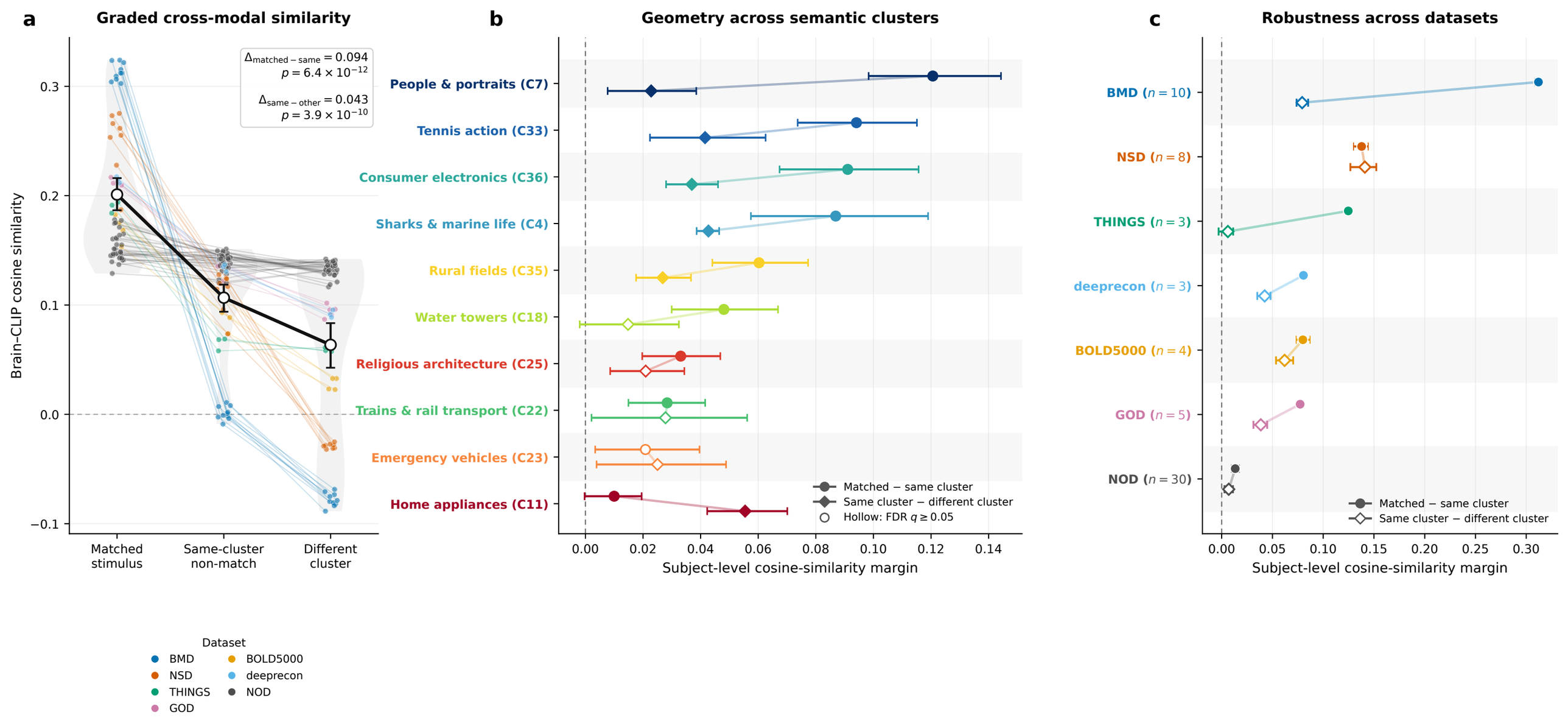

a
Graded cross-modal similarity
$\Delta_{matched-same} = 0.094$
$p = 6.4 \times 10^{-12}$
$\Delta_{same-other} = 0.043$
$p = 3.9 \times 10^{-10}$
Brain–CLIP cosine similarity
0.3
0.2
0.1
0.0
−0.1
Matched stimulus
Same-cluster non-match
Different cluster
Dataset
BMD
NSD
THINGS
GOD
BOLD5000
deeprecon
NOD
b
Geometry across semantic clusters
People & portraits (C7)
Tennis action (C33)
Consumer electronics (C36)
Sharks & marine life (C4)
Rural fields (C35)
Water towers (C18)
Religious architecture (C25)
Trains & rail transport (C22)
Emergency vehicles (C23)
Home appliances (C11)
Matched − same cluster
Same cluster − different cluster
Hollow: FDR $q \geq 0.05$
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
Subject-level cosine-similarity margin
c
Robustness across datasets
BMD ($n = 10$)
NSD ($n = 8$)
THINGS ($n = 3$)
deeprecon ($n = 3$)
BOLD5000 ($n = 4$)
GOD ($n = 5$)
NOD ($n = 30$)
Matched − same cluster
Same cluster − different cluster
0.00
0.05
0.10
0.15
0.20
0.25
0.30
Subject-level cosine-similarity margin